Chapter 2

Vector Spaces

The operations of addition and scalar multiplication are used in many diverse contexts in mathematics. The system of this form are called vector spaces or linear spaces. In this chapter the definition of a vector space is given and some of the general theory of vector spaces is given.

We will present the formal definition of a vector space. Before doing this, we begin with taking a look at the Euclidean vector space $\mathbb{R}^n$. Let $\mathbb{R}$ be the set of all real numbers and $\mathbb{R}^n$ be the set of all n-vectors with real components, that is,

$$\mathbb{R}^n = \left\{ \vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \;\middle|\; u_i \in \mathbb{R} \right\}.$$

Scalar multiplication and addition in $\mathbb{R}^n$ are defined by

$$\alpha\vec{u} = \begin{bmatrix} \alpha u_1 \\ \alpha u_2 \\ \vdots \\ \alpha u_n \end{bmatrix} \text{ and } \vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}$$

for any $\vec{u}, \vec{v} \in \mathbb{R}^n$ and any scalar $\alpha \in \mathbb{R}$.

Example 2.0.1

Let $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \in \mathbb{R}^3$. Then $\vec{u} + \vec{v} = \begin{bmatrix} 1+0 \\ 0+1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and $3\vec{u} = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$.

The above discussion leads us to the following theorem for the space $\mathbb{R}^n$ which motivates us to define a vector space in general.

Theorem 2.0.1

If $\vec{u}, \vec{v}$ and $\vec{w} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$, then

1. $\vec{u} + \vec{v} \in \mathbb{R}^n$
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4. there exists a zero vector $\vec{0} \in \mathbb{R}^n$ such that $\vec{u} + \vec{0} = \vec{u} = \vec{0} + \vec{u}$
5. there exists an n-vector $-\vec{u} \in \mathbb{R}^n$ such that $\vec{u} + (-\vec{u}) = \vec{0} = +(-\vec{u}) + \vec{u}$
6. $\alpha\vec{u} \in \mathbb{R}^n$
7. $(\alpha\beta)\vec{u} = \alpha(\beta\vec{u})$
8. $(\alpha + \beta)\vec{u} = \alpha\vec{u} + \beta\vec{u}$
9. $\alpha(\vec{u} + \vec{v}) = \alpha\vec{u} + \alpha\vec{v}$
10. $1\vec{u} = \vec{u}$.

More generally, let $\mathbb{R}^{m\times n}$ denote the set of all $m \times n$ matrices with real entries. If $A = (a_{ij})$ and $B = (b_{ij})$, the sum $A + B$ is defined to be $(a_{ij} + b_{ij})$. Given a scalar $\alpha \in \mathbb{R}$, we define αA to be (αa_{ij}). These operations also obey certain arithmetic rules similarly to the case of $\mathbb{R}^n$. Therefore, these rules form the axioms that are used to define the concept of a vector space.

Definition 2.0.1

A vector space V over $\mathbb{R}$ or $\mathbb{C}$ is a nonempty set with addition and scalar multiplication satisfying the following conditions for all $\vec{u}, \vec{v}$ and $\vec{w} \in V$ and $\alpha, \beta \in \mathbb{R}$ or $\mathbb{C}$.

1. $\vec{u} + \vec{v} \in V$
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4. there exists a zero vector $\vec{0} \in V$ such that $\vec{u} + \vec{0} = \vec{u} = \vec{0} + \vec{u}$
5. there exists an n-vector $-\vec{u} \in V$ such that $\vec{u} + (-\vec{u}) = \vec{0} = +(-\vec{u}) + \vec{u}$
6. $\alpha\vec{u} \in V$
7. $(\alpha\beta)\vec{u} = \alpha(\beta\vec{u})$
8. $(\alpha + \beta)\vec{u} = \alpha\vec{u} + \beta\vec{u}$
9. $\alpha(\vec{u} + \vec{v}) = \alpha\vec{u} + \alpha\vec{v}$
10. $1\vec{u} = \vec{u}$.

Note that the elements of V are called *vectors* and the real numbers or complex numbers are called *scalar*.

The next two examples present counterexamples of sets which are not vector spaces.

Example 2.0.2: Counterexample

The set $V := \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \,\middle|\, x + y = 1 \right\}$ is <u>not</u> a vector space.

Observe that if $\vec{u} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then both $\vec{u}$ and $\vec{v}$ are in V. However,

$$\vec{u} + \vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin V, \qquad \vec{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \notin V, \qquad -\vec{u} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \notin V, \qquad 2\vec{u} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \notin V.$$

Thus, conditions 1, 4, 5, 6 failed.

Example 2.0.3: Counterexample

Determine whether $V := \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \,\middle|\, x^2 = y^2 \right\}$ is a vector space.

Observe that if $\vec{u} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}$, then both $\vec{u}$ and $\vec{v}$ are in V. However,

$$\vec{u} + \vec{v} = \begin{bmatrix} 5 \\ -1 \end{bmatrix} \notin V.$$

Thus, condition 1 failed. Therefore V is not a vector space.

We show here once again for an example of a general vector space which is not $\mathbb{R}^n$.

Example 2.0.4

Show that $V := \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \,\middle|\, a, b, c, d \in \mathbb{R} \right\}$ is a vector space over $\mathbb{R}$.

All the properties can be verified.

1. $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix} \in V$
2. Clear
3. Clear
4. There exists $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in V$ satisfying the condition.
5. For each $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, there is $-\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \in V$
6. - 10. are clear.

2.1 Subspaces

Many of vector spaces we will consider are subsets of other vector spaces. For example, the set

$$V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x + y = 0 \right\}$$

which can be shown to be a vector space over $\mathbb{R}$ is a subset of $\mathbb{R}^2$. Such vector spaces we shall assign a name for them as follows.

Let V be a vector space and $\emptyset \neq U \subseteq V$. If U is itself a vector space under the same addition and scalar multiplication as V, then U is called a *subspace of* V.

Before presenting examples of subspaces we provide a theorem that makes it relatively easy to determine whether a subset of a linear space V is a subspace of V.

Theorem 2.1.1

A nonempty subset U of a vector space V is a subspace of V if and only if

1. If $\vec{u}, \vec{v} \in U$, then $\vec{u} + \vec{v} \in U$.
2. If $\vec{u} \in U$ and $\alpha \in \mathbb{R}$, then $\alpha\vec{u} \in U$.

The conditions described in this theorem can be restated with the following terminology:

1. U is closed under addition.
2. U is closed under scalar multiplication.

It is worth noting that every vector space contains at least one subspace: the set of a singleton zero. The following example contains less trivial example of subspace.

Example 2.1.1

Consider the subset U of $\mathbb{R}^2$ defined by $U = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \;\middle|\; 2x + 3y = 0 \right\}$ is a subspace of $\mathbb{R}^2$.

Solution:

$U \neq \emptyset$ because $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in U$.

1. Let $\vec{u} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \in U$.

Then $2x_1 + 3y_1 = 0$ and $2x_2 + 3y_2 = 0$.

Consider $\vec{u} + \vec{v} = \begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix}$.

Check: $2(x_1 + x_2) + 3(y_1 + y_2) = 2x_1 + 2x_2 + 3y_1 + 3y_2$

$= (2x_1 + 3y_1) + (2x_2 + 3y_2) = 0 + 0 = 0.$

Therefore $\vec{u} + \vec{v} \in U$.

2. Let $\vec{u} = \begin{bmatrix} x \\ y \end{bmatrix}$ and $\alpha \in \mathbb{R}$. Then $2x + 3y = 0$.

Consider $\alpha\vec{u} = \begin{bmatrix} \alpha x \\ \alpha y \end{bmatrix}$

Check: $2(\alpha x) + 3(\alpha y) = \alpha(2x + 3y) = \alpha \cdot 0 = 0$. Therefore $\alpha\vec{u} \in U$.

Therefore, U is a subspace of $\mathbb{R}^2$.

We can also use the theorem to verify the next counterexample of subspace.

Example 2.1.2: Counterexample

Consider the subset U of $\mathbb{R}^2$ defined by $U = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \;\middle|\; x + y = 1 \right\}$.

Since $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \in U$ but $2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \notin U$, then U is not a subspace of $\mathbb{R}^2$.

2.2 Linear Combinations

This section focuses on the smallest subspace of a vector space V containing a given finite set $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ of elements of V. This subspace contains all scalar products of these elements and all possible sums of these scalar products. Therefore, we shall consider sums of the form

$$\vec{w} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$$

where $c_1, c_2, \ldots, c_n$ are scalars.

Definition 2.2.1

Let V be a vector space and $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n \in V$. An element $\vec{w} \in V$ is called a *linear combination* of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n$ if there are scalars $c_1, c_2, \ldots, c_n \in \mathbb{R}$ such that $\vec{w} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$

Let's take a look at the following examples.

Example 2.2.1

Given $\vec{u} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$.

1. Then $\vec{w} = \begin{bmatrix} -3 \\ 0 \\ -3 \end{bmatrix}$ is a linear combination of $\vec{u}$ and $\vec{v}$, because

$$\begin{bmatrix} -3 \\ 0 \\ -3 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + (-3) \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, \quad \text{that is,} \quad \vec{w} = 6\vec{u} + (-3)\vec{v}.$$

2. Consider $\vec{z} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Show that $\vec{z}$ is not a linear combination of $\vec{u}$ and $\vec{v}$.

Suppose $\vec{z} = x_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, that is, $\begin{aligned} x_1 + 3x_2 &= 1 \\ x_1 + 2x_2 &= 1 \\ 2x_1 + 5x_2 &= 1 \end{aligned}$.

$$\left[\begin{array}{cc|c} 1 & 3 & 1 \\ 1 & 2 & 1 \\ 2 & 5 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & -1 & 0 \\ 0 & -1 & -1 \end{array}\right] \begin{array}{l} R_2 + (-R_1) \\ R_3 + (-2R_1) \end{array} \sim \left[\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{array}\right] R_3 + (-R_2).$$

No solution, that is, No x_1, x_2 such that $x_1\vec{u} + x_2\vec{v} = \vec{z}$.

Example 2.2.2

Given $\vec{u} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \vec{v} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$. Write $\vec{b} = \begin{bmatrix} 1 \\ -7 \\ -4 \end{bmatrix}$ as a linear combination of $\vec{u}$ an $\vec{v}$.

Solution: $\vec{b} = x_1\vec{u} + x_2\vec{v} \iff \begin{aligned} 2x_1 + x_2 &= 1 \\ x_1 + 3x_2 &= -7 \\ x_1 + 2x_2 &= -4 \end{aligned}$

$$\left[\begin{array}{cc|c} 2 & 1 & 1 \\ 1 & 3 & -7 \\ 1 & 2 & -4 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & -4 \\ 1 & 3 & -7 \\ 2 & 1 & 1 \end{array}\right] \begin{matrix} R_{13} \\ \\ \\ \end{matrix} \sim \left[\begin{array}{cc|c} 1 & 2 & -4 \\ 0 & 1 & -3 \\ 0 & -3 & 9 \end{array}\right] \begin{matrix} \\ R_2 + (-R_1) \\ R_3 + (-2R_1) \end{matrix}$$

$$\sim \left[\begin{array}{cc|c} 1 & 2 & -4 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right] \begin{matrix} \\ \\ R_3 + 3R_2 \end{matrix} \sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right] \quad R_1 + (-2R_2)$$

Therefore, $\left.\begin{aligned} x_1 &= 2 \\ x_2 &= -3 \end{aligned}\right\} \implies \vec{b} = 2\vec{u} + (-3)\vec{v}.$

The equation in variables $x_1, x_2, \ldots, x_n$ of form $x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_n\vec{v}_n = \vec{b}$ is called *vector equation*, it corresponds to the augmented matrix $\left[\vec{v}_1 \;\; \vec{v}_2 \;\; \cdots \;\; \vec{v}_n \,\middle|\, \vec{b}\right]$.

The following example shows a relation between the concept of matrix equation and linear combination of vectors.

Example 2.2.3

Find two vectors $\vec{v}_1$ and $\vec{v}_2$ such that every solution of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \vec{x} = \vec{0}$ is a linear combination of $\vec{v}_1$ and $\vec{v}_2$.

Solution: $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ R_2 + (-2R_1) \\ R_3 + (-3R_1) \end{matrix}$,

that is, $x_1 + 2x_2 + 3x_3 = 0 \implies x_1 = -2x_2 - 3x_3$ where x_2, x_3 are free.

The solution $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 - 3x_3 \\ x_2 + 0x_3 \\ 0x_2 + x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \\ 0x_2 \end{bmatrix} + \begin{bmatrix} -3x_3 \\ 0x_3 \\ x_3 \end{bmatrix}$

$= x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$.

Therefore, $\vec{v}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$.

2.3 Spanning Sets

Let V be a vector space over $\mathbb{R}$. Let $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p \in V$. We call the set of all linear combination of $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$ *the subset of* V *spanned by* $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p$, denoted by

$$\text{Span}\left\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\right\} = \left\{c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p \;\middle|\; c_1, c_2, \ldots, c_p \in \mathbb{R}\right\}.$$

Remark: $\text{Span}\left\{\vec{0}\right\} = \left\{\vec{0}\right\}$.

If $\text{Span}\left\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\right\} = V$, that is, every vector in V can be expressed as a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$, we say that $\vec{v}_1, \ldots, \vec{v}_p$ span V. It is straightforward that $\text{Span}\left\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\right\}$ is a subspace of V and we record it in the following theorem.

Theorem 2.3.1

If $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p \in V$, then $\text{Span}\left\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\right\}$ is a subspace of V.

Example 2.3.1

1. $\text{Span}\left\{\begin{bmatrix}2\\1\\3\end{bmatrix}\right\} = \left\{c\begin{bmatrix}2\\1\\3\end{bmatrix} \;\middle|\; c \in \mathbb{R}\right\}$.

Observe that $\begin{bmatrix}4\\2\\6\end{bmatrix} \in \text{Span}\left\{\begin{bmatrix}2\\1\\3\end{bmatrix}\right\}$.

2. $\text{Span}\left\{\begin{bmatrix}2\\1\\1\end{bmatrix}, \begin{bmatrix}1\\3\\2\end{bmatrix}\right\} = \left\{c_1\begin{bmatrix}2\\1\\1\end{bmatrix} + c_2\begin{bmatrix}1\\3\\2\end{bmatrix} \;\middle|\; c_1, c_2 \in \mathbb{R}\right\}$.

Observe that $\begin{bmatrix}1\\-7\\-4\end{bmatrix} \in \text{Span}\left\{\begin{bmatrix}2\\1\\1\end{bmatrix}, \begin{bmatrix}1\\3\\2\end{bmatrix}\right\}$ since $\begin{bmatrix}1\\-7\\-4\end{bmatrix} = 2\begin{bmatrix}2\\1\\1\end{bmatrix} + (-3)\begin{bmatrix}1\\3\\2\end{bmatrix}$.

The following example presents a well-known spanning set of $\mathbb{R}^3$.

Example 2.3.2

Let $\vec{e}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}$, $\vec{e}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix} \in \mathbb{R}^3$. For any $\vec{v} = \begin{bmatrix}a\\b\\c\end{bmatrix} \in \mathbb{R}^3$, $\vec{v} = a\vec{e}_1 + b\vec{e}_2 + c\vec{e}_3$. Indeed, $\begin{bmatrix}a\\b\\c\end{bmatrix} = a\begin{bmatrix}1\\0\\0\end{bmatrix} + b\begin{bmatrix}0\\1\\0\end{bmatrix} + c\begin{bmatrix}0\\0\\1\end{bmatrix}$. Therefore, $\mathbb{R}^3 = \text{Span}\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$.

Note that the similar concept can be generalized to the case of $\mathbb{R}^n$.

The following examples show that a spanning set problem can be associated to the problem of solving a system of equations.

Example 2.3.3

Given $\vec{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix}$. Determine whether $\vec{b} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$ is an element of Span$\{\vec{v}_1, \vec{v}_2\}$ or not ?

Solution: If $\vec{b} \in$Span$\{\vec{v}_1, \vec{v}_2\}$, then $\vec{b} = x_1\vec{v}_1 + x_2\vec{v}_2$, where $x_1, x_2 \in \mathbb{R}$. Thus,

$$x_1 \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} \quad \Longrightarrow \quad \begin{aligned} -x_1 + 2x_2 &= 1 \\ 2x_1 - 3x_2 &= 1 \\ 3x_1 + 5x_2 &= -3 \end{aligned}$$

$$\left[\begin{array}{cc|c} -1 & 2 & 1 \\ 2 & -3 & 1 \\ 3 & 5 & -3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & 1 & 3 \\ 0 & 11 & 0 \end{array}\right] \begin{array}{c} R_1 \\ R_2 + 2R_1 \\ R_3 + 3R_1 \end{array}$$

$$\sim \left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & -33 \end{array}\right] \quad R_3 + (-11R_2)$$

No solution. Therefore, $\vec{b} \notin \text{Span}\{\vec{v}_1, \vec{v}_2\}$.

Example 2.3.4

Find the value of h for which $\vec{w} = \begin{bmatrix} 5 \\ 9 \\ h \end{bmatrix}$ is an element of Span$\{\vec{v}_1, \vec{v}_2\}$, where

$\vec{v}_1 = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$.

Solution: $\vec{w} = x_1\vec{v}_1 + x_2\vec{v}_2$.

$$\begin{aligned} 5 &= x_1 + 2x_2 \\ 9 &= -3x_1 + 6x_2 \\ h &= 2x_1 + 4x_2 \end{aligned} \quad \longleftrightarrow \quad \left[\begin{array}{cc|c} 1 & 2 & 5 \\ -3 & 6 & 9 \\ 2 & 4 & h \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 12 & 24 \\ 0 & 0 & h-10 \end{array}\right] \begin{array}{c} \\ R_2 + 3R_1 \\ R_3 - 2R_1 \end{array}$$

The system is consistent if and only if $h - 10 = 0$. Therefore, $h = 10$.

2.4 Linearly Independence

In this section, we introduce the concepts of linear dependence and linear independence. These simple concepts provide the keys to understanding the structure of vector spaces.

Let V be a vector space over $\mathbb{R}$ and $S \subseteq V$. We say that S is *linearly independent* if for every vector $\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p \in S$ and $c_1, c_2, \ldots, c_p \in \mathbb{R}$,

$$\text{if } \; c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_p\vec{v}_p = \vec{0}, \quad \text{then } c_1 = c_2 = \cdots = c_p = 0.$$

We say that S is *linearly dependent* if otherwise.

Let's try to understand this concept by using several examples.

Example 2.4.1

Let $S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \subseteq \mathbb{R}^2$.

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Longrightarrow \quad \begin{matrix} c_1 = 0 \\ c_2 = 0 \end{matrix}.$$

Therefore S is linear independent.

Example 2.4.2

Given $\vec{v}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$. $\vec{v}_2 = \begin{bmatrix} -3 \\ -5 \\ 2 \end{bmatrix}$ and $\vec{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. Determine whether $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is linear independent.

Solution: $c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$

$$\begin{bmatrix} 1 & -3 & 0 \\ 3 & -5 & 1 \\ -2 & 2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & 0 \\ 0 & 4 & 1 \\ 0 & -4 & -1 \end{bmatrix} \begin{matrix} \\ R_2 - 3R_1 \\ R_3 + 2R_1 \end{matrix} \sim \begin{bmatrix} 1 & -3 & 0 \\ 0 & 4 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ R_3 + R_2 \end{matrix}$$

$$\begin{aligned} c_1 - 3c_2 &= 0 \\ 4c_2 + c_3 &= 0 \end{aligned} \quad \Longrightarrow \quad \begin{aligned} c_1 &= 3c_2 \\ c_3 &= -4c_2 \\ c_2 &\text{ free variable.} \end{aligned}$$

Therefore, $\vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 3c_2 \\ c_2 \\ -4c_2 \end{bmatrix} = c_2 \begin{bmatrix} 3 \\ 1 \\ -4 \end{bmatrix}$.

We may choose $c_1 = 3,\ c_2 = 1,\ c_3 = -4$. Then $3\vec{v}_1 + \vec{v}_2 - 4\vec{v}_3 = 0$.

Therefore, $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is linear dependent.

Example 2.4.3

Let $\mathbb{R}^{\mathbb{R}}$ be a vector space of all functions from $\mathbb{R}$ to $\mathbb{R}$. Consider $S = \{e^x,\ \sin x\}$.

$$c_1 e^x + c_2 \sin x = 0, \quad \text{for all } x \in \mathbb{R}$$

Set $x = 0$; $\quad c_1 e^0 + c_2 \sin 0 = 0 \quad \Longrightarrow \quad c_1 = 0.$

Set $x = \dfrac{\pi}{2}$; $\quad 0 \cdot e^{\pi/2} + c_2 \sin \pi/2 = 0 \quad \Longrightarrow \quad c_2 = 0.$

Therefore, $S = \{e^x,\ \sin x\}$ is linearly independent.

Example 2.4.4

Find the value of h for which $\left\{ \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ h \end{bmatrix} \right\}$ is linearly independent.

Solution:

$$c_1 \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 5 \\ h \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This system must have only one trivial solution, which is $c_1 = c_2 = c_3 = 0$.

$$\begin{bmatrix} 1 & 3 & -1 \\ -1 & -5 & 5 \\ 4 & 7 & h \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 \\ 0 & -2 & 4 \\ 0 & -5 & h+4 \end{bmatrix} \begin{matrix} R_2 + R_1 \\ R_3 - 4R_1 \end{matrix} \sim \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & -5 & h+4 \end{bmatrix} -\tfrac{1}{2}R_2$$

$$\sim \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & h-6 \end{bmatrix} R_3 + 5R_2 \qquad \cdots \sim \cdots \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\implies \begin{matrix} h - 6 \neq 0 \\ h \neq 6. \end{matrix}$$

Example 2.4.5

Let $\vec{v}_1 = \begin{bmatrix}1\\4\\7\end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix}2\\5\\8\end{bmatrix}$, $\vec{v}_3 = \begin{bmatrix}3\\6\\9\end{bmatrix}$. Determine whether $S = \{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is independent. If it is linearly dependent, then write $\vec{v}_1$ as a linear combination of $\vec{v}_2$ and $\vec{v}_3$.

Solution:

$$\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix} \sim \begin{bmatrix}1&2&3\\0&-3&-6\\0&-6&-12\end{bmatrix} \begin{matrix}\\R_2-4R_1\\R_3-7R_1\end{matrix} \sim \begin{bmatrix}1&2&3\\0&1&2\\0&0&0\end{bmatrix} \begin{matrix}\\-\frac{1}{3}R_2\\R_3-2R_2\end{matrix}$$

$$\begin{matrix}c_1+2c_2+3c_3=0\\c_2+2c_3=0\end{matrix} \implies \begin{matrix}c_2=-2c_3\\c_1-4c_3+3c_3=0\end{matrix}$$

$$\implies \begin{matrix}c_1=c_3\\c_2=-2c_3\end{matrix} \text{ where } c_3 \text{ is free.}$$

Thus $\vec{c} = \begin{bmatrix}c_1\\c_2\\c_3\end{bmatrix} = \begin{bmatrix}c_3\\-2c_3\\c_3\end{bmatrix} = c_3\begin{bmatrix}1\\-2\\1\end{bmatrix}$. Therefore, S is linearly dependent.

Moreover, we may choose $c_1 = 1,\ c_2 = -2,\ c_3 = 1$.

Thus $\vec{v}_1 - 2\vec{v}_2 + \vec{v}_3 = \vec{0}$. Therefore, $\vec{v}_1 = 2\vec{v}_2 - \vec{v}_3$.

Example 2.4.6

Consider $\mathbb{R}[x]$ the vector space of all polynomials over $\mathbb{R}$. Is $S = \{x, x^2, 1-x\}$ linearly independent ?

$$c_1 x + c_2 x^2 + c_3(1-x) = 0, \qquad \forall x \in \mathbb{R}.$$

$$c_3 + (c_1 - c_3)x + c_2 x^2 = 0 \quad (*)$$

$$x = 0 \quad \Longrightarrow \quad c_3 = 0$$

$$(*); \qquad c_1 x + c_2 x^2 = 0, \quad \forall x \in \mathbb{R}.$$

$$x = 1 \quad \Longrightarrow \quad c_1 + c_2 = 0$$

$$x = -1 \quad \Longrightarrow \quad -c_1 + c_2 = 0$$

$$\Longrightarrow \quad 2c_2 = 0 \quad \Longrightarrow \quad c_2 = 0 \quad \Longrightarrow \quad c_1 = 0.$$

Therefore, the set S is linearly independent.

2.5 Basis

In the previous section, one can realize that a spanning set for a vector space is minimal if its elements are linearly independent. The elements of a minimal spanning set form the basic building blocks for the whole vector space, and consequently, we say that form a *basis* for the vector space.

Definition 2.5.1

Let V be a vector space over $\mathbb{R}$. A finite set $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} \subseteq V$ is called a *basis* of V if

1. S is linearly independent, and
2. S spans V, that is, Span$S = V$.

The next example presents a well-known basis of $\mathbb{R}^3$.

Example 2.5.1

In $\mathbb{R}^3$, $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis of $\mathbb{R}^3$

Remark: $\left\{ \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a standard basis of $\mathbb{R}^3$.

Example 2.5.2

Is $S = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ a basis for $\mathbb{R}^2$?

Solution:

1. $c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$; $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 \\ 0 & -3 \end{bmatrix}$ $R_2 - 2R_1$

   $$\implies \begin{array}{l} c_1 + 2c_2 = 0 \\ -3c_2 = 0 \end{array}.$$

   Therefore, $c_1 = c_2 = 0$ (linearly independent).

2. Let $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$.

   $$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}.$$

   $$\left[\begin{array}{cc|c} 1 & 2 & a \\ 2 & 1 & b \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & a \\ 0 & -3 & b - 2a \end{array}\right] \quad R_2 - 2R_1$$

   $$\begin{array}{l} x_1 + 2x_2 = a \\ -3x_2 = b - 2a \end{array} \implies x_2 = \frac{2a - b}{3}$$

   $$\implies x_1 = a - 2x_2 = a - 2\left(\frac{2a - b}{3}\right) = \frac{-a + 2b}{3}.$$

   Thus $\begin{bmatrix} a \\ b \end{bmatrix} = \frac{-a + 2b}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2a - b}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, that is,

   $\begin{bmatrix} a \\ b \end{bmatrix}$ can be expressed as a linear combination of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

   Thus S spans $\mathbb{R}^2$. Therefore, S is a basis for $\mathbb{R}^2$.

For a vector space $\mathbb{R}^n$, there are many equivalent properties for S being a basis.

Theorem 2.5.1

Let $S = \{\vec{v}_1, \ldots, \vec{v}_n\} \subseteq \mathbb{R}^n$, Let A be the $n \times n$ matrix whose column are $\vec{v}_1, \ldots, \vec{v}_n$, that is, $A = \begin{bmatrix} | & | & & | \\ \vec{v}_1 & \vec{v}_2 & \ldots & \vec{v}_n \\ | & | & & | \end{bmatrix}$. The following statements are equivalent:

1. S is a basis $\mathbb{R}^n$;
2. S is linearly independent;
3. S spans $\mathbb{R}^n$;
4. A is nonsingular, that is, $\det A \neq 0$;
5. $A\vec{x} = \vec{0}$ has $\vec{x} = \vec{0}$ as it only solution;
6. $A\vec{x} = \vec{b}$ is consistent for any $\vec{b} \in \mathbb{R}^n$.

Example 2.5.3

Is $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$ a basis for $\mathbb{R}^3$?

Solution: $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$. Consider $A\vec{x} = 0$. Then

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \quad R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad R_3 + R_2 \qquad \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \tfrac{1}{2}R_3$$

$$\left.\begin{aligned} x_1 + x_2 &= 0 \\ x_2 + x_3 &= 0 \\ x_3 &= 0 \end{aligned}\right\} \quad \Longrightarrow \quad x_1 = x_2 = x_3 = 0.$$

Thus $\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is the only solution.

Therefore, S is a basis for $\mathbb{R}^3$.

Example 2.5.4

Let V be a vector space of all 2×2 matrices over $\mathbb{R}$. Is $S := \left\{\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}, \begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}, \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}, \begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\right\}$ a basis for V ?

Solution: 1. Check if S is linearly independent.

$$c_1\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix} + c_2\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix} + c_3\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} + c_4\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix} \implies \begin{aligned} c_1 + 2c_2 \quad\quad + c_4 &= 0\\ c_1 + c_2 + c_3 + c_4 &= 0\\ c_2 + c_3 + c_4 &= 0\\ c_1 + c_2 \quad\quad\quad &= 0\end{aligned}$$

$$\begin{bmatrix}1 & 2 & 0 & 1\\ 1 & 1 & 1 & 1\\ 0 & 1 & 1 & 1\\ 1 & 1 & 0 & 0\end{bmatrix} \sim \begin{bmatrix}1 & 2 & 0 & 1\\ 0 & -1 & 1 & 0\\ 0 & 1 & 1 & 1\\ 0 & -1 & 0 & -1\end{bmatrix} \begin{matrix}\\ R_2 - R_1\\ \\ R_4 - R_1\end{matrix} \sim \begin{bmatrix}1 & 2 & 0 & 1\\ 0 & -1 & 1 & 0\\ 0 & 0 & 2 & 1\\ 0 & 0 & -1 & -1\end{bmatrix} \begin{matrix}\\ \\ R_3 + R_2\\ R_4 - R_2\end{matrix}$$

$$\sim \begin{bmatrix}1 & 2 & 0 & 1\\ 0 & -1 & 1 & 0\\ 0 & 0 & 2 & 1\\ 0 & 0 & 0 & -\frac{1}{2}\end{bmatrix} \begin{matrix}\\ \\ \\ R_4 + \frac{1}{2}R_3\end{matrix} \implies \left.\begin{aligned} c_1 + 2c_2 \quad + c_4 &= 0\\ -c_2 + c_3 \quad &= 0\\ 2c_3 + c_4 &= 0\\ -\tfrac{1}{2}c_4 &= 0\end{aligned}\right\} \begin{matrix} c_1 = c_2 = c_3 = c_4 = 0\\ S \text{ is linearly independent}\end{matrix}$$

2. To show that S spans V. Let $\begin{bmatrix}a & b\\ c & d\end{bmatrix} \in V$. Then

$$c_1\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix} + c_2\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix} + c_3\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} + c_4\begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}a & b\\ c & d\end{bmatrix} \implies \begin{aligned} c_1 + 2c_2 \quad\quad + c_4 &= a\\ c_1 + c_2 + c_3 + c_4 &= b\\ c_2 + c_3 + c_4 &= c\\ c_1 + c_2 \quad\quad\quad &= d\end{aligned}$$

We can apply the same sequence of row operation of 1. (above) to obtain a solution (c_1, c_2, c_3, c_4). Therefore, S spans V.

Find a basis of U (a subspace of V)

Let $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ be a set of nonzero vectors of V and let U be the subspace of V spanned by S, that is, $U = \text{Span}S$. From S, we can find a basis for U as follows:

1. Let $S_1 = \{\vec{v}_1\}$. Then S_1 is linearly independent.

2. $S_2' = S_1 \cup \{\vec{v}_2\}$ $\begin{cases} S_2 = S_2' & \text{if} \quad S_2' \text{ is linearly independent} \\ S_2 = S_1 & \text{if} \quad \text{otherwise} \end{cases}$
   Therefore, S_2 is linearly independent and $\text{Span}S_2 = \text{Span}\{\vec{v}_1, \vec{v}_2\}$.

3. $S_3' = S_2 \cup \{\vec{v}_3\}$ $\begin{cases} S_3 = S_3' & \text{if} \quad S_3' \text{ is linearly independent} \\ S_3 = S_2 & \text{if} \quad \text{otherwise} \end{cases}$
   Therefore, S_3 is linearly independent and $\text{Span}S_3 = \text{Span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$.

4. Continuing in this manner we obtain a set that is linearly independent and spans U. Hence, this set is a basis of U.

Example 2.5.5

Let $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$. Consider $U = \text{Span}S$ a subspace of $\mathbb{R}^4$. Find a basis for U.

Solution:

Let $S_1 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$, $S_2' = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ $c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \implies c_1 = c_2 = 0$.

Therefore S_2' is linearly independent $\implies$ $S_2 = S_2'$.

Let $S_3' = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}$ $3 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

Therefore S_3' is linearly independent $\implies$ $S_3 = S_2$.

Let $S_4' = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$ $c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \longleftrightarrow$

$$\begin{aligned} c_1 + \quad c_3 &= 0 \\ c_2 + c_3 &= 0 \\ c_1 + c_2 \quad &= 0 \end{aligned}.$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \underset{R_3 - R_1}{\sim} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \underset{R_3 - R_2}{\sim} \begin{matrix} c_1 + c_3 = 0 \\ c_2 = 0 \\ -c_3 = 0 \end{matrix} \Bigg\} \begin{matrix} c_1 = 0 \\ c_2 = 0 \\ c_3 = 0 \end{matrix}$$

Therefore S_4' is linearly independent $\implies$ $S_4 = S_4'$ and so S_4 is a basis for U.

Example 2.5.6

Let $A = \begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ -2 & 3 & 7 \end{bmatrix}$. Consider the solution space of $A\vec{x} = \vec{0}$, that is, $U = \left\{ \vec{x} \in \mathbb{R}^3 \;\middle|\; A\vec{x} = \vec{0} \right\}$. Find a basis for U.

Solution:

$$\begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ -2 & 3 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ 0 & -3 & -3 \end{bmatrix} \underset{R_3 + 2R_1}{\sim} \begin{bmatrix} 1 & -3 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \underset{R_3 + 3R_2}{\sim} \begin{cases} x_1 - 3x_2 - 2x_3 = 0 \\ x_2 - x_3 = 0 \end{cases}$$

$$\begin{array}{l} x_2 = x_3 \\ x_1 = 5x_3 \\ x_3 \text{ free} \end{array} \longleftrightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5x_3 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}. \quad U = \left\{ x_3 \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} \;\middle|\; x_3 \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Since $\left\{ \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} \right\}$ is linearly independent, $\left\{ \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix} \right\}$ is a basis for U.

Example 2.5.7

Find a basis for the subspace $U = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \;\middle|\; a + b = 0,\ b + c = 0 \right\}$ of $\mathbb{R}^3$

Solution:

$$\begin{array}{rcl} a + b = 0 & \Longrightarrow & a = -b \\ b + c = 0 & \Longrightarrow & c = -b \end{array}, \; b \in \mathbb{R} \text{ free. Thus}$$

$$U = \left\{ \begin{bmatrix} -b \\ b \\ -b \end{bmatrix} \;\middle|\; b \in \mathbb{R} \right\} = \left\{ b \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \;\middle|\; b \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\}$$

Therefore, $\left\{ \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\}$ is a basis for U.

Example 2.5.8

Find a basis for the subspace $U = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \;\middle|\; a + 2b + 3c + d = 0 \right\}$ of $\mathbb{R}^4$.

Solution: $a + 2b + 3c + d = 0 \implies a = -2b - 3c - d$. Any vector in U is of the form

$$\begin{bmatrix} -2b - 3c - d \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} -2b \\ b \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -3c \\ 0 \\ c \\ 0 \end{bmatrix} + \begin{bmatrix} -d \\ 0 \\ 0 \\ d \end{bmatrix} = b \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

$$U = \text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Observe that $c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

$\implies$ $c_1 = c_2 = c_3 = 0$.So, the set is linearly independent.

Therefore, $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for U.

2.5.1 Coordinate Vectors

Let S be a basis for a linear space V. Since S spans V, any element $\vec{v}$ of V can be written as a linear combination of the elements of S. The following theorem is the foundation for the development of any further theory of linear spaces.

Theorem 2.5.2

Let V be a vector space over $\mathbb{R}$. Let $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ be a basis for V. Let $\vec{v} \in V$. Then there are unique scalars $c_1, c_2, \ldots, c_n$ such that $\vec{v} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$.

Fix a basis $\mathcal{B} = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ for a vector space V. By ordered basis we mean that $\mathcal{B}$ is a basis for V and the order in which the elements of $\mathcal{B}$ are written is fixed.

If $\vec{v} \in V$, then by the above theorem there are unique numbers $c_1, c_2, \ldots, c_n$ such that $\vec{v} = c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$. Since the basis $\mathcal{B}$ is ordered we can represent the linear combination $c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_n\vec{v}_n$ by the n-vector which called *the coordinate vector of $\vec{v}$ relative to $\mathcal{B}$*,

$$[\vec{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}.$$

Conversely, given an n-vector

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

we can associate a unique element $b_1v_1 + b_2v_2 + \cdots + b_nv_n$ of V. Hence, every vector space (with the real numbers as scalars) having a basis with n elements can be identified with $\mathbb{R}^n$.

Example 2.5.9

Let $\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 2 \\ 3 \\ -4 \\ 1 \end{bmatrix}$, and $\vec{x} = \begin{bmatrix} 0 \\ -5 \\ 6 \\ -3 \end{bmatrix}$. Fix $\mathcal{B} = \{\vec{v}_1, \vec{v}_2\}$. ($\mathcal{B}$ is a basis for $\text{Span}\{\vec{v}_1, \vec{v}_2\}$). Check if $\vec{x} \in \text{Span}\{\vec{v}_1, \vec{v}_2\}$ or not. If so, find $[\vec{x}]_{\mathcal{B}}$.

Solution: $\vec{x} = c_1\vec{v}_1 + c_2\vec{v}_2$

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ -1 & 3 & -5 \\ 1 & -4 & 6 \\ -1 & 1 & -3 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 5 & -5 \\ 0 & -6 & 6 \\ 0 & 3 & -3 \end{array}\right] \begin{array}{l} \\ R_2 + R_1 \\ R_3 - R_1 \\ R_4 + R_1 \end{array}$$

$$\sim \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{array}\right] \begin{array}{l} \\ \frac{1}{5}R_2 \\ \frac{1}{6}R_3 \\ \frac{1}{3}R_4 \end{array} \sim \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ \\ R_3 + R_2 \\ R_4 - R_2 \end{array}$$

$\left.\begin{array}{r} c_1 + 2c_2 = 0 \\ c_2 = -1 \end{array}\right\}$ $c_1 = 2$. Therefore, $\vec{x} = 2\vec{v}_1 - \vec{v}_2$. So, $[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

2.5.2 Dimension

Note that a vector space can have more than one basis. In this section, we present one of the unifying theorems of linear algebra theory: Any two bases of V have the same number of elements. The number of elements n in a basis for V is a magic number, because if S is a subset of V with n elements, then S is linear independent if and only if S spans V. Hence, knowing that a set has the right number of elements allows us to determine whether the set is a basis by verifying only one of the two properties required for the set to be a basis.

Let V be a vector space over $\mathbb{R}$.

Theorem 2.5.3

Suppose that $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is a basis for V. Then every basis for V has n elements.

Definition 2.5.2

Let $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ be a basis for V. Then V is called a *finite dimensional vector space* and is said to have *dimension n*, denoted by $\dim V = n$. A vector space that does not have finite dimension is called an *infinite dimensional vector space*.

Remark: The $\left\{\vec{0}\right\}$ is a vector space over $\mathbb{R}$ having dimension zero. Indeed, the basis for $\left\{\vec{0}\right\}$ is $\emptyset$.

In this course, we only focus on finite dimensional vector space.

Example 2.5.10

In $\mathbb{R}^3$, the set $\left\{\vec{e}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \vec{e}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \vec{e}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ is a basis for $\mathbb{R}^3$.

Then $\dim\mathbb{R}^3 = 3$.

In general, $\dim\mathbb{R}^n = n$.

Theorem 2.5.4

Let U be a subspace of V. Then $\dim U \leq \dim V$.

In general, to show that a set $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is a basis for a vector space V we must show that S is both linearly independent and spans V. However, the following theorem allows us to only verify that S has one of these properties.

Theorem 2.5.5

Let $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} \subseteq V$ and $\dim V = n$. Then the following statements are equivalent.

1. S is a basis for V.
2. S is linearly independent.
3. S spans V.

Theorem 2.5.6

Let $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\}$ be a linearly independent subset V and $\dim V = n$ and $k < n$. Then there are elements $\vec{v}_{k+1}, \ldots, \vec{v}_n \in V$ such that $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ is a basis for V.

Example 2.5.11

Consider the linearly independent subset $S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ of $\mathbb{R}^4$. Find a basis of $\mathbb{R}^4$ containing S.

Solution:

Set $\mathcal{B}_0 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \vec{e}_1, \vec{e}_2, \vec{e}_3, \vec{e}_4 \right\}$ $\quad \text{Span}\mathcal{B}_0 = \mathbb{R}^4$.

Look at $\vec{e}_4$; it cannot expressed as a linear combination of the remaining vector.

Set $\mathcal{B}_1 = \mathcal{B}_0$.

Consider $\vec{e}_3$. $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$. Therefore delete $\vec{e}_3$ from $\mathcal{B}_1$.

Set $\mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. Consider $\vec{e}_2$. $\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

Therefore delete $\vec{e}_2$ from $\mathcal{B}_2$.

Set $\mathcal{B}_3 = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$. Since $\text{Span}\mathcal{B}_3 = \mathbb{R}^4$, $\mathcal{B}_3$ is a basis for $\mathbb{R}^4$.

Theorem 2.5.7

Let V be an n-dimentional vector space. Let $S = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_k\} \subseteq V$.

1. If $k < n$, then Span$S \neq V$.
2. If $k > n$, then S is linearly dependent.

Example 2.5.12

Find the dimension of the subspace $U = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \;\middle|\; a + b + c = 0 \right\}$ of $\mathbb{R}^3$.

Solution: $a + b + c = 0 \quad \Longrightarrow \quad a = -b - c,\ b, c \in \mathbb{R}$.

$$U = \left\{ \begin{bmatrix} -b - c \\ b \\ c \end{bmatrix} \;\middle|\; b, c \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -b \\ b \\ 0 \end{bmatrix} + \begin{bmatrix} -c \\ 0 \\ c \end{bmatrix} \;\middle|\; b, c \in \mathbb{R} \right\}$$

$$= \left\{ b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \;\middle|\; b, c \in \mathbb{R} \right\}.$$

Thus $U = \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

Note that $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ is linearly independent, so it is a basis for U.

Therefore, dim$U = 2$.

2.5.3 Change of basis

In some applications involving vector spaces it is advantageous to use a basis other than the one given.

Suppose that $\mathcal{B}$ is a basis for a vector space V and that we wish to use another basis $\mathcal{B}'$. That is, we wish to write elements of V as linear combinations of the elements of $\mathcal{B}'$ rather than as linear combinations of the elements of $\mathcal{B}$.

Theorem 2.5.8

Let V be an n-dimentional vector space. Let $\mathcal{B} = \{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ and $\mathcal{B}' = \{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_n\}$ be ordered bases for V. Let ${}_{\mathcal{B}'}P_{\mathcal{B}} = \begin{bmatrix} | & | & & | \\ [\vec{v}_1]_{\mathcal{B}'} & [\vec{v}_2]_{\mathcal{B}'} & \cdots & [\vec{v}_n]_{\mathcal{B}'} \\ | & | & & | \end{bmatrix}$. Then ${}_{\mathcal{B}'}P_{\mathcal{B}}$ is nonsingular and $[\vec{v}]_{\mathcal{B}'} = {}_{\mathcal{B}'}P_{\mathcal{B}}\,[\vec{v}]_{\mathcal{B}}$ for every $\vec{v} \in V$.

The matrix ${}_{\mathcal{B}'}P_{\mathcal{B}}$ is called the *transition matrix* from Then $\mathcal{B}$ to Then $\mathcal{B}'$.

Example 2.5.13

Consider $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ and $\mathcal{B}' = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$ for $\mathbb{R}^2$. Find ${}_{\mathcal{B}'}P_{\mathcal{B}}$ and ${}_{\mathcal{B}}P_{\mathcal{B}'}$.

Solution: Write $\vec{e}_1$ and $\vec{e}_2$ in term of linear combination of $\vec{v}_1$ and $\vec{v}_2$, that is,

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}; \qquad \begin{bmatrix} 0 \\ 1 \end{bmatrix} = d_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + d_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

From short calculations, we have $c_1 = -\frac{1}{3},\ c_2 = \frac{2}{3},\ d_1 = \frac{2}{3},\ d_2 = -\frac{1}{3}$.

Therefore $[\vec{e}_1]_{\mathcal{B}'} = \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$, $[\vec{e}_2]_{\mathcal{B}'} = \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \end{bmatrix}$, and so, ${}_{\mathcal{B}'}P_{\mathcal{B}} = \begin{bmatrix} -\frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$.

Similarly, $\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}; \qquad \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$

Therefore $[\vec{v}_1]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $[\vec{v}_2]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and so, ${}_{\mathcal{B}}P_{\mathcal{B}'} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

Remark: If ${}_{\mathcal{B}'}P_{\mathcal{B}} \cdot {}_{\mathcal{B}}P_{\mathcal{B}'} = I_n$, then $({}_{\mathcal{B}'}P_{\mathcal{B}})^{-1} = {}_{\mathcal{B}}P_{\mathcal{B}'}$.

Example 2.5.14

Given that $\mathcal{B} = \left\{\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix}\right\}$, $\mathcal{B}' = \left\{\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\1\end{bmatrix}\right\}$, $\mathcal{N} = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$ are bases for $\mathbb{R}^2$. Find ${}_{\mathcal{N}}P_{\mathcal{B}}$, ${}_{\mathcal{B}'}P_{\mathcal{N}}$, ${}_{\mathcal{B}'}P_{\mathcal{B}}$.

Answer: ${}_{\mathcal{N}}P_{\mathcal{B}} = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$ and ${}_{\mathcal{N}}P_{\mathcal{B}'} = \begin{bmatrix}1 & 2\\2 & 1\end{bmatrix}$

${}_{\mathcal{B}'}P_{\mathcal{N}} = ({}_{\mathcal{N}}P_{\mathcal{B}'})^{-1} = \begin{bmatrix}1 & 2\\2 & 1\end{bmatrix}^{-1} = \begin{bmatrix}-\frac{1}{3} & \frac{2}{3}\\\frac{2}{3} & -\frac{1}{3}\end{bmatrix}$.

Moreover, ${}_{\mathcal{B}'}P_{\mathcal{B}} = \begin{bmatrix}\frac{1}{3} & -1\\\frac{1}{3} & 1\end{bmatrix}$.

Remark: Note that ${}_{\mathcal{B}'}P_{\mathcal{B}} = ({}_{\mathcal{B}'}P_{\mathcal{N}})\,({}_{\mathcal{N}}P_{\mathcal{B}})$.

2.6 Row Space, Column Space and Rank of Matrix

If A is an $m \times n$ matrix, each row of A is an n-tuples of real numbers and hence can be considered as a vector in $R^{1\times n}$. The m vectors corresponding to the rows of A will be referred to as the row vectors of A. Similarly, each column of A can be considered as a vector in $\mathbb{R}^m$ and one can associate n column vectors with A.

Definition 2.6.1: Row Space and Column Space

Let A be an $m \times n$ matrix. The vector space spanned by the rows of A is called the *row space of* A, denoted by $\text{Row}(A)$. The vector space spanned by the columns of A is called the *column space of* A, denoted by $\text{Col}(A)$.

Example 2.6.1

Let $A = \begin{bmatrix}1 & 0 & 2\\0 & 1 & 0\\0 & 1 & 1\end{bmatrix}$. Then $\text{Row}(A) = \text{Span}\left\{\begin{bmatrix}1 & 0 & 2\end{bmatrix}, \begin{bmatrix}0 & 1 & 0\end{bmatrix}, \begin{bmatrix}0 & 1 & 1\end{bmatrix}\right\}$.

$\text{Col}(A) = \text{Span}\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}2\\0\\1\end{bmatrix}\right\}$.

We have the relationship between the solutions of the matrix equation $A\vec{x} = \vec{b}$ and the column space of A.

Theorem 2.6.1

Let A be an $m \times n$ matrix. The equation $A\vec{x} = \vec{b}$ is consistent if and only if $\vec{b} \in \text{Col}(A)$.

This theorem clearly indicates the importance of being able to determine the column space of a matrix A.

Definition 2.6.2

Two matrices A and B are said to be *row equivalent* if A can be changed into B by using elementary row operations.

Theorem 2.6.2

Suppose that A is row equivalent to B that is in row echelon form.

1. The nonzero rows of $\mathcal{B}$ form a basis for $\text{Row}(A)$.
2. Let $A_1, A_2, \ldots, A_n$ be the columns of A and $B_1, B_2, \ldots, B_n$ be the columns of B. If $B_{i_1}, B_{i_2}, \ldots, B_{i_k}$ are the columns of B containing a nonzero leading entry of some row of B, then the corresponding columns $A_{i_1}, A_{i_2}, \ldots, A_{i_k}$ form a basis for $\text{Col}(A)$.

Example 2.6.2

Given $A = \begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \\ 3 & 3 & 1 & 4 & 4 & 3 \\ 1 & 1 & 1 & 1 & 2 & -1 \\ 3 & 3 & 0 & 3 & 3 & 3 \end{bmatrix}$. By using elementary row operations, the matrix A is equivalent to $B = \begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

$\text{Row}(A)$ has a basis $\left\{\begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & -1 & 0 & -2 \end{bmatrix}\right\}$.

$\text{Col}(A)$ has a basis $\left\{\begin{bmatrix} 1 \\ 3 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 1 \\ 3 \end{bmatrix}\right\}$.

Theorem 2.6.3

Let A be any matrix. Then $\dim(\text{Row}(A)) = \dim(\text{Col}(A))$.

Definition 2.6.3

For a matrix A, rank of A ($\text{rank}(A)$) is equal to $\dim(\text{Col}(A))$.

Remark: $\text{rank}(A)$ = number of nonzero leading entries of an echelon form of A.

Theorem 2.6.4

The equation $A\vec{x} = \vec{b}$ is consistent if and only is $\text{rank}(A) = \text{rank}([A|b])$.

Fix an $m \times n$ matrix A. The solution space at $A\vec{x} = \vec{0}$, $\text{Nul}(A) = \left\{\vec{x} \mid A\vec{x} = \vec{0}\right\}$, is called the *null space* of A. The dimension of $\text{Nul}(A)$ is called the *nullity of* A, denoted by $\text{nullity}(A)$.

Theorem 2.6.5

Let A be an $m \times n$ matrix. Then $\text{rank}(A) + \text{nullity}(A) = n$.

Example 2.6.3

Let $A = \begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \\ 3 & 3 & 1 & 4 & 4 & 3 \\ 1 & 1 & 1 & 1 & 2 & -1 \\ 3 & 3 & 0 & 3 & 3 & 3 \end{bmatrix}$. Find $\text{rank}(A)$ and $\text{nullity}(A)$.

Solution: Known A is row equivalent to $B = \begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$. Therefore,

$\text{rank}(A) = 3$.

$\text{rank}(A)+\text{nullity}(A) = 6 \implies 3+\text{nullity}(A) = 6.$ Therefore, $\text{nullity}(A) = 3$.

Verify:

$$\begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 2 & 2 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ -R_3 \\ \\ \end{matrix} \sim \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 + x_2 \quad\quad + x_5 - x_6 &= 0 \\ x_3 \quad + x_5 - 2x_6 &= 0 \\ x_4 \quad + 2x_6 &= 0 \end{aligned} \implies \begin{cases} x_1 = -x_2 - x_5 + x_6 \\ x_3 = -x_5 + 2x_6 \\ x_4 = -2x_6, \qquad x_2, x_5, x_6 \in \mathbb{R}. \end{cases}$$

Therefore, any $\vec{x} \in \text{Nul}(A)$ is

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -x_2 - x_5 + x_6 \\ x_2 \\ -x_5 + 2x_6 \\ -2x_6 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -x_5 \\ 0 \\ -x_5 \\ 0 \\ x_5 \\ 0 \end{bmatrix} + \begin{bmatrix} x_6 \\ 0 \\ 2x_6 \\ -2x_6 \\ 0 \\ x_6 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_6 \begin{bmatrix} 1 \\ 0 \\ 2 \\ -2 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore, $\text{Nul}(A)$ has $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right\}$ as a basis. Thus $\text{nullity}(A) = 3$.

2.7 Orthogonality

One can add to the structure of a vector space by defining a scalar or inner product. This will enable us to describe geometric properties such as length, angle, and orthogonality in this vector space. Here we mainly focus on vector spaces having the real numbers as scalars.

For $\vec{u}, \vec{v} \in \mathbb{R}^n$, the product $\vec{u}^t\vec{v}$ is called *inner product* or *scalar product* of vectors $\vec{u}$ and $\vec{v}$, denoted by $\vec{u} \cdot \vec{v}$.

Suppose $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$, $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$. Then $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \cdots + u_nv_n \in \mathbb{R}$.

It is immediate to have the following theorem.

Theorem 2.7.1

For $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^n$ and $c \in \mathbb{R}$,

1. $\vec{u} \cdot \vec{0}_n = 0$
2. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
3. $(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}$
4. $(c\vec{u}) \cdot \vec{v} = c(\vec{u} \cdot \vec{v}) = \vec{u} \cdot (c\vec{v})$
5. $\vec{u} \cdot \vec{u} \geq 0$ and $\vec{u} \cdot \vec{u} = 0$ if and only if $\vec{u} = \vec{0}$.

Length or **Norm** of a vector $\vec{v} = (v_1, v_2, \ldots, v_n)$ denoted $\|\vec{v}\|$ is

$$\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

Remark: $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$ and $\|c\vec{v}\| = |c|\|\vec{v}\|$ for all $c \in \mathbb{R}$.

Distance between vectors $\vec{x}$ and $\vec{y}$ is $\|\vec{x} - \vec{y}\|$.

The unit vector (length 1) with the same direction to $\vec{v}$ is $\dfrac{\vec{v}}{\|\vec{v}\|}$, provided $\vec{v} \neq \vec{0}$.

Example 2.7.1

Given $\vec{v} = (1, 0, 2, -2) \in \mathbb{R}^4$. Find the unit vector with the same direction to $\vec{v}$, and the vector with length 3 and the opposite direction to $\vec{v}$.

Solution: $\|\vec{v}\| = \sqrt{1^2 + 0^2 + 2^2 + (-2)^2} = \sqrt{9} = 3$.

- The unit vector with the same direction to $\vec{v}$ is $\dfrac{\vec{v}}{\|\vec{v}\|} = \dfrac{1}{3}\begin{bmatrix} 1 \\ 0 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 0 \\ 2/3 \\ -2/3 \end{bmatrix}$.
- The vector with length 3 and the opposite direction to $\vec{v}$ is $\dfrac{-3\vec{v}}{\|\vec{v}\|} = -3\begin{bmatrix} 1/3 \\ 0 \\ 2/3 \\ -2/3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ -2 \\ 2 \end{bmatrix}$.

In $\mathbb{R}^2$,it is not difficult to show that the angle between two line segments is a right angle if and only if the inner product of the corresponding vectors in zero. In general, if V is a vector space with an inner product, then two vectors in V are said orthogonal will be defined as follows.

For $\vec{u}, \vec{v} \in \mathbb{R}^n$, we say that $\vec{u}$ and $\vec{v}$ are **orthogonal** (or perpendicular) if $\vec{u} \cdot \vec{v} = 0$.

Remark One can think of orthogonality as a generalization of the concept of perpendicular to any vector space with an inner product.

We call a set of nonzero vectors $S = \{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p\} \subseteq \mathbb{R}^n$ an orthogonal set if $\vec{u}_i \cdot \vec{u}_j = 0$ for all $i \neq j$.

Example 2.7.2

Determine if $S = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \right\}$ is an orthogonal set in $\mathbb{R}^3$.

Solution:

$\vec{v}_1 \cdot \vec{v}_2 = 1 \cdot 0 + (-2) \cdot 1 + 1 \cdot 2 = 0 \quad \vec{v}_1 \cdot \vec{v}_3 = 1 \cdot (-5) + (-2) \cdot (-2) + 1 \cdot 1 = 0$

$\vec{v}_2 \cdot \vec{v}_3 = 0 \cdot (-5) + 1 \cdot (-2) + 2(1) = 0 \quad$ Therefore, S is an orthogonal set in $\mathbb{R}^3$.

Theorem 2.7.2

If $S = \{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p\}$ is an orthogonal set in $\mathbb{R}^n$, then S is linearly independent. Hence, S is a basis for $H = \text{Span}\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p\}$.

A basis $\mathcal{B}$ for a subspace H of $\mathbb{R}^n$ is called an *orthogonal basis* if $\mathcal{B}$ is an orthogonal set.

Remark: In $\mathbb{R}^n$, $\mathcal{B} = \{\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n\}$ is an orthogonal basis.

Let $\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p\}$ be an orthogonal basis for a subspace H of $\mathbb{R}^n$. Want to express any $\vec{y} \in H$ as a linear combination of $\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p$.

Theorem 2.7.3

Under the above setup.

$$\vec{y} = \frac{\vec{y} \cdot \vec{u}_1}{\|\vec{u}_1\|^2}\vec{u}_1 + \frac{\vec{y} \cdot \vec{u}_2}{\|\vec{u}_2\|^2}\vec{u}_2 + \cdots + + \frac{\vec{y} \cdot \vec{u}_p}{\|\vec{u}_p\|^2}\vec{u}_2.$$

Example 2.7.3

Consider $S = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \right\}$ is an orthogonal basis for $\mathbb{R}^3$. Write $\vec{y} = \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix}$ as a linear combination of all element in S.

Solution: Let $\vec{u}_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $\vec{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $\vec{u}_3 = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix}$. Then

$$\vec{y} = \frac{\vec{y} \cdot \vec{u}_1}{\|\vec{u}_1\|^2}\vec{u}_1 + \frac{\vec{y} \cdot \vec{u}_2}{\|\vec{u}_2\|^2}\vec{u}_2 + \cdots + \frac{\vec{y} \cdot \vec{u}_p}{\|\vec{u}_p\|^2}\vec{u}_2 = \frac{6}{6}\vec{u}_1 + \frac{1}{5}\vec{u}_2 + \left(-\frac{2}{5}\right)\vec{u}_3 = \vec{u}_1 + \frac{1}{5}\vec{u}_2 - \frac{2}{5}\vec{u}_3.$$

An orthogonal set (basis) containing unit vectors is called an *orthonormal set* (basis).

Example 2.7.4

$S' = \left\{ \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \frac{1}{\sqrt{30}} \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix} \right\}$ is an orthonormal basis for $\mathbb{R}^3$.

We can observe that

Theorem 2.7.4

Let U be an $m \times n$ matrix. Then the set of all columns of U is orthonormal if and only if $U^t U = I_n$.

Theorem 2.7.5

Let U be an $m \times n$ matrix whose the set of all columns of U is orthonormal. Then

1. $U\vec{x} \cdot U\vec{y} = \vec{x} \cdot \vec{y}$
2. $\|U\vec{x}\| = \|\vec{x}\|$
3. $U\vec{x} \cdot U\vec{y} = 0 \quad \text{iff} \quad \vec{x} \cdot \vec{y} = 0.$

If the set of all column of U is orthogonal (orthonormal), then U is called an orthogonal matrix (orthonomal matrix).

Remark: U is orthonormal matrix iff U is invertible and $U^{-1} = U^t$.

2.8 Projection and Gram-Schmidt Process

Let $\vec{z}$ be a vector in $\mathbb{R}^n$ and H be a subspace of $\mathbb{R}^n$, we say that $\vec{z}$ is *orthogonal* to H if $\vec{z}$ is orthogonal to all vectors in H. We call the set of vectors orthogonal to H the **orthogonal complement** of H, denoted by $H^\perp$ (sometimes called "H perp"), that is,

$$H^\perp = \left\{\vec{z} \in \mathbb{R}^n \;\middle|\; \vec{z} \cdot \vec{a} = 0 \text{ for all } \vec{a} \in H\right\}$$

Remark: $H \cap H^\perp = \left\{\vec{0}\right\}$.

Theorem 2.8.1

Let H be a subspace of $\mathbb{R}^n$. Then

1. $H^\perp$ is a subspace of $\mathbb{R}^n$.
2. If $H = \text{Span}\left\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p\right\}$, then $\vec{z} \in H^\perp$ iff $\vec{z} \cdot \vec{u}_1 = \vec{z} \cdot \vec{u}_2 = \cdots = \vec{z} \cdot \vec{u}_p = 0$.

Remark: If $H = \text{Span}\left\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p\right\}$, then

$$\begin{aligned} H^\perp &= \left\{\vec{z} \in \mathbb{R}^n \;\middle|\; \vec{z} \cdot \vec{u}_1 = \vec{z} \cdot \vec{u}_2 = \cdots = \vec{z} \cdot \vec{u}_p = 0\right\} \\ &= \left\{\vec{z} \in \mathbb{R}^n \;\middle|\; \vec{u}_1^t\vec{z} = \vec{u}_2^t\vec{z} = \cdots = \vec{u}_p^t\vec{z} = 0\right\} \\ &= \text{Nul}(A^t), \end{aligned}$$

where $A = \begin{bmatrix} | & | & & | \\ \vec{u}_1 & \vec{u}_2 & \cdots & \vec{u}_p \\ | & | & & | \end{bmatrix}$.

Example 2.8.1

Given $H = \text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}\right\}$. Find a basis for $H^\perp$.

Solution: $H^\perp = \text{Nul}(A^t)$

$$A = \begin{bmatrix}1 & 0\\1 & 1\\0 & 1\end{bmatrix}. \quad A^t = \begin{bmatrix}1 & 1 & 0\\0 & 1 & 1\end{bmatrix} \sim \begin{bmatrix}1 & 0 & -1\\0 & 1 & 1\end{bmatrix} \quad \begin{matrix} x_1 - x_3 = 0 \\ x_2 + x_3 = 0 \end{matrix} \Bigg\} \begin{matrix} x_1 = x_3 \\ x_2 = -x_3 \\ x_3 \in \mathbb{R} \end{matrix}.$$

$$\text{Therefore } \vec{x} \in \text{Nul}(A^t) \implies \vec{x} = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}x_3\\-x_3\\x_3\end{bmatrix} = x_3\begin{bmatrix}1\\-1\\1\end{bmatrix}.$$

Therefore, $H^\perp = \text{Nul}(A^t) = \text{Span}\left\{\begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}$. So, a basis for $H^\perp$ is $\left\{\begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}$.

Let $\vec{u} \in \mathbb{R}^n$ be a nonzero vector, let $H = \text{Span}\{\vec{u}\}$. Then any $\vec{y} \in \mathbb{R}^n$, $\vec{y} = \hat{y} + \vec{z}$, where $\hat{y} = \alpha\vec{u}$ for some $\alpha \in \mathbb{R}$ and $\vec{z} \perp \vec{u}$. Find α =?

$$\vec{z} = \vec{y} - \hat{y} = \vec{y} - \alpha\vec{u} \perp \vec{u}. \quad \text{So, } (\vec{y} - \alpha\vec{u}) \cdot \vec{u} = 0 \implies \vec{y} \cdot \vec{u} - \alpha\,(\vec{u} \cdot \vec{u}) = 0.$$

Therefore, $\alpha = \dfrac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}$ and $\hat{y} = \left(\dfrac{\vec{y} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}\right)\vec{u}$.

In general, we have

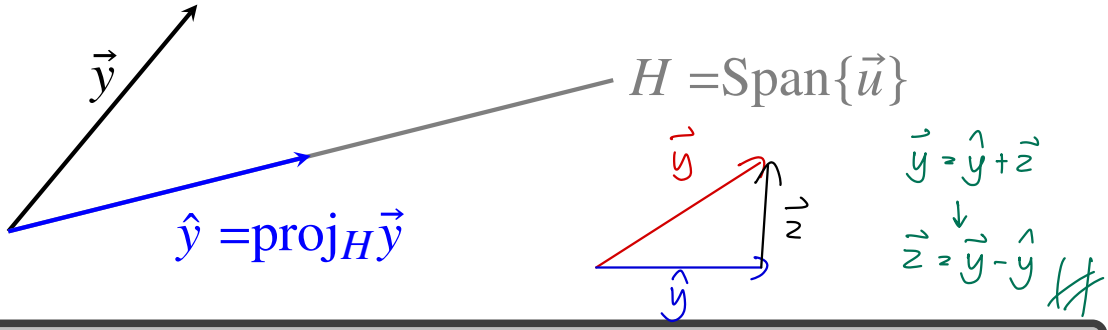
Theorem 2.8.2: Orthogonal Decomposition

Let $\{\vec{u}_1, \ldots, \vec{u}_p\}$ be an orthogonal set in $\mathbb{R}^n$, where $\vec{u}_i \neq 0$ and $H = \text{Span}\{\vec{u}_1, \ldots, \vec{u}_p\}$. Then any $\vec{y} \in \mathbb{R}^n$,

$$\hat{y} = \frac{\vec{y} \cdot \vec{u}_1}{\vec{u}_1 \cdot \vec{u}_1}\vec{u}_1 + \frac{\vec{y} \cdot \vec{u}_2}{\vec{u}_2 \cdot \vec{u}_2}\vec{u}_2 + \cdots + \frac{\vec{y} \cdot \vec{u}_p}{\vec{u}_p \cdot \vec{u}_p}\vec{u}_p \quad \text{and} \quad \vec{z} = \vec{y} - \hat{y} \in H^\perp$$

such that $\vec{y} = \hat{y} + \vec{z}$.

Remark: $\hat{y}$ is called the orthogonal projection of $\vec{y}$ onto H, denoted by $\text{proj}_H\vec{y}$ and $\vec{z}$ is called *complement* of $\vec{y}$ perpendicular to H.



Example 2.8.2

Given $\vec{u}_1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $\vec{u}_2 = \begin{bmatrix}-1\\4\\1\end{bmatrix}$, $\vec{u}_3 = \begin{bmatrix}2\\1\\-2\end{bmatrix}$, and $\vec{x} = \begin{bmatrix}8\\-4\\-3\end{bmatrix}$. Write $\vec{x}$ as a sum of $\vec{y}$ and $\vec{z}$, where $\vec{y} \in$Span$\{\vec{u}_1, \vec{u}_2\}$ and $\vec{z} =$Span$\{\vec{u}_3\}$.

Solution:

$$\left.\begin{aligned}\vec{u}_1 \cdot \vec{u}_2 = 0\\ \vec{u}_1 \cdot \vec{u}_3 = 0\\ \vec{u}_2 \cdot \vec{u}_3 = 0\end{aligned}\right\} \quad \Longrightarrow \quad \{\vec{u}_1, \vec{u}_2\vec{u}_3\} \text{ is an orthogonal set in } \mathbb{R}^3.$$

So, it is an orthonormal basis for $\mathbb{R}^3$. Therefore,

$$\hat{x} = \frac{\vec{x}\cdot\vec{u}_1}{\vec{u}_1\cdot\vec{u}_1}\vec{u}_1 + \frac{\vec{x}\cdot\vec{u}_2}{\vec{u}_2\cdot\vec{u}_2}\vec{u}_2 + \frac{\vec{x}\cdot\vec{u}_3}{\vec{u}_3\cdot\vec{u}_3}\vec{u}_3 = \frac{5}{2}\vec{u}_1 - \frac{3}{2}\vec{u}_2 + 2\vec{u}_3.$$

Therefore $\vec{x} = \vec{y} + \vec{z}$, where $\vec{y} = \frac{5}{2}\vec{u}_1 - \frac{3}{2}\vec{u}_2$ and $\vec{z} = 2\vec{u}_3$.
Moreover, $\vec{y} =$proj$_H\vec{x}$, where $H =$Span$\{\vec{u}_1, \vec{u}_2\}$ and $\vec{z} = \vec{x} - \vec{y} \in$Span$\{\vec{u}_3\} = H^\perp$.

Question: If H is a subspace of $\mathbb{R}^n$, then can we find an orthonormal basis for H?
Answer: Yes, but how?

To answer the question, we consider the following discussion.

If we have a basis $\{\vec{x}_1, \vec{x}_2, \ldots \vec{x}_p\}$ of a subspace H, we proceed to find an orthogonal basis of H by

1. letting $\vec{v}_1 = \vec{x}_1$ and $\vec{v}_2$ to be $\text{proj}_{\text{Span}\{\vec{v}_1\}}\vec{x}_2$
2. lettin $\vec{v}_3$ to be $\text{proj}_{\text{Span}\{\vec{v}_1,\vec{v}_2\}}\vec{x}_3$ and so on.

Then the set $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_p\}$ is an orthogonal basis for H.

We call this process of constructing an orthogonal basis for H as **Gram-Schmidt process** and record it in the following theorem.

Theorem 2.8.3: Gram-Schmidt process

Let $\{\vec{x}_1, \vec{x_2}, \dots \vec{x}_p\}$ be a basis for a subspace H of $\mathbb{R}^n$. Set

$$\begin{aligned}
\vec{v}_1 &= \vec{x}_1 \\
\vec{v}_2 &= \vec{x}_2 - \frac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1}\vec{v}_1 \\
\vec{v}_3 &= \vec{x}_3 - \frac{\vec{x}_3 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1}\vec{v}_1 - \frac{\vec{x}_3 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2}\vec{v}_2 \\
&\vdots \\
\vec{v}_p &= \vec{x}_p - \frac{\vec{x}_p \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1}\vec{v}_1 - \frac{\vec{x}_p \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2}\vec{v}_2 - \cdots - \frac{\vec{x}_p \cdot \vec{v}_{p-1}}{\vec{v}_{p-1} \cdot \vec{v}_{p-1}}\vec{v}_{p-1}.
\end{aligned}$$

Then $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_p\}$ is an orthogonal basis for H.

By normalization of vectors, we have

Corollary 2.8.4

If $\mathcal{B}$ is a basis of a subspace H, then we can always construct an orthonormal basis for H from the basis $\mathcal{B}$.

Example 2.8.3

Construct an orthonormal basis for $H = \text{Span}\left\{ \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\3\end{bmatrix} \right\} \subseteq \mathbb{R}^3$.

Solution: Check if $\left\{ \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\3\end{bmatrix} \right\} \subseteq \mathbb{R}^3$ is a basis for H.

$$A = \begin{bmatrix}0 & -1 & -1\\1 & 0 & 1\\2 & 1 & 3\end{bmatrix} \sim \begin{bmatrix}1 & 0 & 1\\0 & 1 & 1\\0 & 0 & 0\end{bmatrix}.$$

By Theorem, a basis for H is $\left\{ \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix} \right\}$. Let $\vec{x}_1 = \begin{bmatrix}0\\1\\2\end{bmatrix}$ and $\vec{x}_2 = \begin{bmatrix}-1\\0\\1\end{bmatrix}$.

Then $\vec{v}_1 = \vec{x}_1$, $\vec{x}_2 \cdot \vec{v}_1 = 2$ and $\vec{v}_1 \cdot \vec{v}_1 = 5$.

Therefore, $\vec{v}_1 = \begin{bmatrix}0\\1\\2\end{bmatrix}$, $\vec{v}_2 = \vec{x}_2 - \dfrac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1}\vec{v}_1 = \begin{bmatrix}-1\\0\\1\end{bmatrix} - \dfrac{2}{5}\begin{bmatrix}0\\1\\2\end{bmatrix} = \begin{bmatrix}-1\\-2/5\\1/5\end{bmatrix}$.

Thus $\left\{ \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}-1\\-2/5\\1/5\end{bmatrix} \right\}$ is an orthogonal basis for H. We may choose $\left\{ \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}-5\\-2\\1\end{bmatrix} \right\}$ as an orthogonal basis for H.

$$\left\| \begin{bmatrix}0\\1\\2\end{bmatrix} \right\| = \sqrt{0^1 + 1^2 + 2^2} = \sqrt{5} \qquad \left\| \begin{bmatrix}-5\\-2\\1\end{bmatrix} \right\| = \sqrt{(-5)^2 + (-2)^2 + 1^2} = \sqrt{30}.$$

Answer: $\left\{ \dfrac{1}{\sqrt{5}}\begin{bmatrix}0\\1\\2\end{bmatrix}, \dfrac{1}{\sqrt{30}}\begin{bmatrix}-5\\-2\\1\end{bmatrix} \right\}$.

Example 2.8.4

Find an orthogonal basis for $H = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$.

Solution:

$$c_1 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \Longrightarrow \quad \begin{cases} c_1 = 0 \\ \\ c_2 = 0 \end{cases}$$

Therefore, $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is linearly independent. Sonit is a basis for H.

Let $\vec{x}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $\vec{x}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$. Set $\vec{v}_1 = \vec{x}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. $\quad \vec{v}_2 = \vec{x}_2 - \dfrac{\vec{x}_2 \cdot \vec{v}_1}{\vec{v}_1 \cdot \vec{v}_1} \vec{v}_1$.

$\vec{x}_2 \cdot \vec{v}_1 = 1 \cdot 0 + (-1)(1) + 0(-1) = -1 \quad$ and $\quad \vec{v}_1 \cdot \vec{v}_1 = 1^2 + (-1)^2 + 0 = 2$.

Therefore, $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} - \dfrac{(-1)}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + \dfrac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \end{bmatrix}$.

So, $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \end{bmatrix} \right\}$ is an orthogonal basis for H.

Remark: Given $A = \begin{bmatrix} | & | & & | \\ \vec{x}_1 & \vec{x}_2 & \cdots & \vec{x}_n \\ | & | & & | \end{bmatrix}$ Suppose $\{\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_n\}$ is linearly independent. We may use Gram-Schmidt Process to find an orthonormal basis $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$ for $\text{Col}(A)$ because $\vec{v}_i \perp \text{Span}\{\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_{i-1}\}$ for all $i \in \{2, \cdots, n\}$. Thus, $\vec{v}_i \cdot \vec{x}_j = 0$ for all $j < i$.

2.9 Exercises 2

1. Consider the subset U of $\mathbb{R}^2$ defined by

$$U = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 2x + 3y = 0 \right\}.$$

Show that U is a subspace of $\mathbb{R}^2$.

2. Verify whether each of the following sets are linearly independent:

a) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right\}$.

b) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \right\}$.

3. Find the values of a for which the following set is linearly independent:

$$\left\{ \begin{bmatrix} 0 \\ 1 \\ a \end{bmatrix}, \begin{bmatrix} a \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ a \\ 1 \end{bmatrix} \right\}.$$

4. Consider the following set of vectors in $\mathbb{R}^3$:

$$X = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Show that one of the two vectors, $u = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, belong to $Span(X)$, while the other does not. Find a real number α such that $\begin{bmatrix} 1 \\ 1 \\ \alpha \end{bmatrix} \in Span(X)$.

5. Show that the following set of vectors is linearly independent:

$$\left\{ x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, x_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ \vdots \\ 0 \end{bmatrix}, \ldots, x_{n-1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \\ -1 \end{bmatrix} \right\}.$$

   Show that $u = [n-1, -1, -1, ..., -1]^t$ is spanned by the above set of vectors.

6. Find a basis and the dimension of the subspaces S.

$$S = \left\{ x \in \mathbb{R}^3 : x = \alpha \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ for some } \alpha \in \mathbb{R} \right\}.$$

7. Find a basis and the dimension of the subspaces S.

$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : x_1 - 7x_2 + 10x_3 = 0 \right\}.$$

8. Verify that the following is a linear independent set and extend it to a basis for $\mathbb{R}^4$:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}.$$

9. Find the coordinate vector for $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ relative to the ordered basis

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

10. Compute the rank and nullity of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

1. Consider the subset U of $\mathbb{R}^2$ defined by

$$U = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : 2x + 3y = 0 \right\}.$$

Show that U is a subspace of $\mathbb{R}^2$.

Sol!! 1) พิจารณาว่า U ไม่เป็นเซตว่าง ⟶ ให้ $x=0, y=0$
จะได้ $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in S$, $2(0)+3(0)=0$ เป็นจริง //

2) พิจารณาคุณสมบัติปิดการบวก ⇒ $u, v \in S \rightarrow u+v \in S$
ให้ $u = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}$, $v = \begin{bmatrix} a_2 \\ b_2 \end{bmatrix}$ ซึ่ง $u \in S$ และ $v \in S$
จะได้ $u: 2a_1+3b_1=0$
$v: 2a_2+3b_2=0$
$u+v: 2a_1+3b_1+2a_2+3b_2=0$
$2(a_1+a_2)+3(b_1+b_2)=0$
$u+v = \begin{bmatrix} a_1+a_2 \\ b_1+b_2 \end{bmatrix} \in S$ เป็นจริง //

3) พิจารณาคุณสมบัติปิดการคูณ ⇒ ให้ $c \in \mathbb{R}$ แล้ว $cU \in S$
จะได้ $cU = \begin{bmatrix} ca \\ cb \end{bmatrix}$
↳ $2(ca)+3(cb)=0$
$c(2a+3b)=0$
$0=0$ เป็นจริง //

∴ U เป็นปริภูมิย่อยของ $\mathbb{R}^2$ // Ans.

2. Verify whether each of the following sets are linearly independent:

a) $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}1\\0\\-1\end{bmatrix}, \begin{bmatrix}2\\1\\1\end{bmatrix}\right\}$.

b) $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\9\end{bmatrix}\right\}$.

Sol: a) จาก $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}1\\0\\-1\end{bmatrix}, \begin{bmatrix}2\\1\\1\end{bmatrix}\right\} = \{\vec{u}, \vec{v}, \vec{w}\}$

ให้ $x_1\vec{u} + x_2\vec{v} + x_3\vec{w} = \vec{0}$

จะได้ $x_1\begin{bmatrix}1\\2\\3\end{bmatrix} + x_2\begin{bmatrix}1\\0\\-1\end{bmatrix} + x_3\begin{bmatrix}2\\1\\1\end{bmatrix} = 0 \longrightarrow \begin{aligned} x_1 + x_2 + 2x_3 &= 0 \\ 2x_1 \quad\quad + x_3 &= 0 \\ 3x_1 - x_2 + x_3 &= 0 \end{aligned}$

ทำให้ได้ว่า $\begin{bmatrix}1&1&2\\2&0&1\\3&-1&1\end{bmatrix} \sim \begin{bmatrix}1&1&2\\0&-2&-3\\0&-4&-5\end{bmatrix}\begin{matrix}R_2-2R_1\\R_3-3R_1\end{matrix}$

$\sim \begin{bmatrix}1&1&2\\0&2&3\\0&0&1\end{bmatrix}\begin{matrix}-R_2\\R_3-2R_2\end{matrix}$

นั่นคือ $x_1 = x_2 = x_3 = 0$

$\therefore$ ข้อ a) เป็น linearly independent // Ans.

b) จาก $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\9\end{bmatrix}\right\} = \{\vec{u}, \vec{v}, \vec{w}\}$

ให้ $x_1\vec{u} + x_2\vec{v} + x_3\vec{w} = \vec{0}$

จะได้ $x_1\begin{bmatrix}1\\2\\3\end{bmatrix} + x_2\begin{bmatrix}4\\5\\6\end{bmatrix} + x_3\begin{bmatrix}7\\8\\9\end{bmatrix} = 0 \longrightarrow \begin{aligned} x_1 + x_2 + 2x_3 &= 0 \\ 2x_1 \quad\quad + x_3 &= 0 \\ 3x_1 - x_2 + x_3 &= 0 \end{aligned}$

ทำให้ได้ว่า $\begin{bmatrix}1&4&7\\2&5&8\\3&6&9\end{bmatrix} \sim \begin{bmatrix}1&2&3\\0&-3&-6\\0&-6&-12\end{bmatrix}\begin{matrix}R_2-2R_1\\R_3-3R_1\end{matrix}$

$\sim \begin{bmatrix}1&0&-1\\0&1&2\\0&0&0\end{bmatrix}\begin{matrix}R_1-2R_2\\-\frac{1}{3}R_2\end{matrix}$

นั่นคือ $\left.\begin{aligned} x_1 - x_3 &= 0 \\ x_2 + 2x_3 &= 0 \end{aligned}\right\}$ โดยที่ $\begin{aligned} x_1 &= x_3 \\ x_2 &= -2x_3 \\ x_3 &\to \text{free variable} \end{aligned}$; $x_3 \in \mathbb{R}$

จะได้ว่า $x = \begin{bmatrix}x_3\\-2x_3\\x_3\end{bmatrix} = x_3\begin{bmatrix}1\\-2\\1\end{bmatrix}$

$\therefore$ ข้อ b) เป็น linearly dependent // Ans.

(non-linearly independent)

3. Find the values of a for which the following set is linearly independent:

$$\left\{\begin{bmatrix}0\\1\\a\end{bmatrix}, \begin{bmatrix}a\\1\\0\end{bmatrix}, \begin{bmatrix}1\\a\\1\end{bmatrix}\right\}.$$

Sol: จากโจทย์ $\left\{\begin{bmatrix}0\\1\\a\end{bmatrix}, \begin{bmatrix}a\\1\\0\end{bmatrix}, \begin{bmatrix}1\\a\\1\end{bmatrix}\right\} = \{\vec{u},\vec{v},\vec{w}\}$

ให้ $x_1\vec{u} + x_2\vec{v} + x_3\vec{w} = \vec{0}$

จะได้ $x_1\begin{bmatrix}0\\1\\a\end{bmatrix} + x_2\begin{bmatrix}a\\1\\0\end{bmatrix} + x_3\begin{bmatrix}1\\a\\1\end{bmatrix} = 0 \longrightarrow \begin{aligned} ax_2 + x_3 &= 0\\ x_1 + x_2 + ax_3 &= 0\\ ax_1 + x_3 &= 0\end{aligned}$

ทำให้ได้ว่า $\begin{bmatrix}0&a&1\\1&1&a\\a&0&1\end{bmatrix} \sim \begin{bmatrix}1&1&a\\0&a&1\\a&0&1\end{bmatrix} \sim R_2 \leftrightarrow R_1$

$\sim \begin{bmatrix}1&1&a\\0&a&1\\0&-a&1-a^2\end{bmatrix} \sim R_3 - aR_1$

$\sim \begin{bmatrix}1&1&a\\0&a&1\\0&0&2-a^2\end{bmatrix} \sim R_3 + R_2$

จะได้ว่า เซตข้างต้น=เป็น linearly independent , $x_1 = x_2 = x_3 = 0$

เมื่อ $2-a^2 \neq 0$, $a \neq 0$

$\therefore$ $a \neq \pm\sqrt{2}, 0$ // Ans.

4. Consider the following set of vectors in $\mathbb{R}^3$:

$$X = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Show that one of the two vectors, $u = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$, belong to $Span(X)$,

while the other does not. Find a real number α such that $\begin{bmatrix} 1 \\ 1 \\ \alpha \end{bmatrix} \in Span(X)$.

Sol: จากโจทย์ $X = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$

- พิจารณา u

จะได้ $u = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \longrightarrow \begin{array}{r} x_1 - 2x_2 = -1 \\ x_2 = 2 \\ -x_1 + x_2 = 1 \end{array}$

นั่นคือ $\left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & 1 & 2 \\ -1 & 1 & 1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & 1 & 2 \\ 0 & -1 & 0 \end{array}\right] \sim R_3 + R_1$

$\sim \left[\begin{array}{cc|c} 1 & -2 & -1 \\ 0 & -1 & 0 \\ 0 & 1 & 2 \end{array}\right] \sim R_{23}$

$\sim \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{array}\right] \begin{array}{l} \sim R_1 - 2R_2 \\ \sim R_3 + R_2 \end{array}$

$\sim \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{array}\right] \sim -R_2 \longrightarrow$ ไม่มีคำตอบ $\therefore$ u ไม่ได้เป็น $Span(X)$ #

- พิจารณา v

จะได้ $v = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \longrightarrow \begin{array}{r} x_1 - 2x_2 = 0 \\ x_2 = 1 \\ -x_1 + x_2 = -1 \end{array}$

นั่นคือ $\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 1 & 1 \\ -1 & 1 & -1 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{array}\right] \sim R_3 + R_1$

$\sim \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{array}\right] \sim R_{23}$

$\sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{array}\right] \begin{array}{l} \sim R_1 - 2R_2 \\ \sim R_3 + R_2 \end{array}$

$\sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right] \sim -R_2$

ทำให้ได้ว่า $x_1 = 2, x_2 = 1 \longrightarrow v = 2\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$ $\therefore$ v เป็น $Span(X)$ #

- พิจารณา $\begin{bmatrix} 1 \\ 1 \\ \alpha \end{bmatrix}$

จะได้ $x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \alpha \end{bmatrix} \longrightarrow \begin{array}{r} x_1 - 2x_2 = 1 \\ x_2 = 1 \\ -x_1 + x_2 = \alpha \end{array}$

นั่นคือ $\left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 1 & 1 \\ -1 & 1 & \alpha \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & \alpha+1 \end{array}\right] \sim R_3 + R_1$

$\sim \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & -1 & \alpha+1 \\ 0 & 1 & 1 \end{array}\right] \sim R_{23}$

$\sim \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 0 & \alpha+2 \\ 0 & 1 & 1 \end{array}\right] \sim R_2 + R_3$

$\sim \left[\begin{array}{cc|c} 1 & -2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & \alpha+2 \end{array}\right] \sim R_{23}$

ทำให้ได้ว่า $\begin{bmatrix} 1 \\ 1 \\ \alpha \end{bmatrix} \in Span(X)$ เมื่อ $\alpha + 2 = 0 \longrightarrow \therefore \alpha = -2$ #

5. Show that the following set of vectors is linearly independent:

$$\left\{ x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, x_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \\ \vdots \\ 0 \end{bmatrix}, \ldots, x_{n-1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ \vdots \\ -1 \end{bmatrix} \right\}.$$

Show that $u = [n-1, -1, -1, \ldots, -1]^t$ is spanned by the above set of vectors.

Sol!! จากโจทย์ พิจารณาการเป็น spanning set ดังนี้

1) พิจารณา linearly independent

จาก n $\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ -1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{bmatrix}$ ($n-1$) $\sim \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 1 & \cdots & 1 \\ 0 & -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{bmatrix} \overset{R_2+R_1}{\sim} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 1 & \cdots & 1 \\ 0 & 0 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & -1 \end{bmatrix} \overset{R_3+R_2}{\sim} \cdots \sim \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & 1 & 1 & \cdots & 1 \\ 0 & 0 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$ → เมทริกซ์สามเหลี่ยมบน

ทำให้ได้ว่า $c_1 = c_2 = c_3 = \ldots = c_n = 0$

$\therefore$ เป็น linearly independent

2) พิจารณาจาก $u = [n-1, -1, -1, \ldots, -1]^t = \begin{bmatrix} n-1 \\ -1 \\ -1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}$

จะได้ว่า $\left[\begin{array}{ccccc|c} 1 & 1 & 1 & \cdots & 1 & n-1 \\ -1 & 0 & 0 & \cdots & 0 & -1 \\ 0 & -1 & 0 & \cdots & 0 & -1 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & -1 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & 1 & 1 & \cdots & 1 & n-1 \\ 0 & 1 & 1 & \cdots & 1 & n-2 \\ 0 & 0 & 1 & \cdots & 1 & n-3 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & n-n \end{array}\right]$ จะเห็นว่า จากเวกเตอร์ $n-1$ ตัวแปร จะมีทั้งหมด $n-1$ สมการ ทำให้ได้ว่า สมการสามารถหาคำตอบได้ 0

$\therefore$ $u \in \text{span}\{\text{เวกเตอร์ทั้งหมด}\}$ // Ans.

6. Find a basis and the dimension of the subspaces S.

$$S = \left\{ x \in \mathbb{R}^3 : x = \alpha \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \text{ for some } \alpha \in \mathbb{R} \right\}.$$

Sol.ⁿ สากลโลกย์ จะได้ว่า

$\therefore$ basis of S คือ $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$ และ $\dim(S) = 1$ # Ans.

7. Find a basis and the dimension of the subspaces S.

$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : x_1 - 7x_2 + 10x_3 = 0 \right\}.$$

Sol.ⁿ สากลโลกย์ ให้ x_2, x_3 เป็น free variable ; $x_2, x_3 \in \mathbb{R}$

จะได้ $x_1 - 7x_2 + 10x_3 = 0$

$$x_1 = 7x_2 - 10x_3$$

$$x_1 = x_2 \begin{bmatrix} 7 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -10 \\ 0 \\ 1 \end{bmatrix}$$

นั่นคือ $S = \text{Span}\left\{ \begin{bmatrix} 7 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ 0 \\ 1 \end{bmatrix} \right\}$ $\longrightarrow$ ทำให้ได้ว่า เป็น linearly independent ด้วย

$\therefore$ basis of S คือ $\left\{ \begin{bmatrix} 7 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -10 \\ 0 \\ 1 \end{bmatrix} \right\}$ และ $\dim(S) = 2$ # Ans.

8. Verify that the following is a linear independent set and extend it to a basis for $\mathbb{R}^4$:

$$\left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}\right\}.$$

Sol. จากโจทย์ จะพิจารณา ดังนี้

ให้ $x_1\begin{bmatrix}1\\0\\-1\\2\end{bmatrix} + x_2\begin{bmatrix}0\\0\\1\\2\end{bmatrix} = 0 \rightarrow \begin{bmatrix}1&0\\0&0\\-1&1\\2&2\end{bmatrix} \sim \begin{bmatrix}1&0\\0&0\\0&1\\0&2\end{bmatrix} \sim \begin{bmatrix}1&0\\0&0\\0&1\\0&0\end{bmatrix} \sim \begin{bmatrix}1&0\\0&1\\0&0\\0&0\end{bmatrix}$ จะได้ $x_1=0, x_2=0$

$\therefore$ เซตข้างต้นเป็น linearly independent

ให้ $S = \left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}, \underbrace{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}}_{e_1}, \underbrace{\begin{bmatrix}0\\1\\0\\0\end{bmatrix}}_{e_2}, \underbrace{\begin{bmatrix}0\\0\\1\\0\end{bmatrix}}_{e_3}, \underbrace{\begin{bmatrix}0\\0\\0\\1\end{bmatrix}}_{e_4}\right\}$

จะเห็นว่า e_2 เป็น basis เนื่องจากไม่มีผลรวมเชิงเส้นใดๆ ที่ได้เป็น นั่นคือ $S_1 = \left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}\right\}$

- พิจารณา $S_2' = \left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\\0\end{bmatrix}\right\}$

จะได้ว่า $c_1\begin{bmatrix}1\\0\\-1\\2\end{bmatrix} + c_2\begin{bmatrix}0\\0\\1\\2\end{bmatrix} + c_3\begin{bmatrix}0\\1\\0\\0\end{bmatrix} + c_4\begin{bmatrix}1\\0\\0\\0\end{bmatrix} = 0 \rightarrow \begin{bmatrix}1&0&0&1&0\\0&0&1&0&0\\-1&1&0&0&0\\2&2&0&0&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0\\0&0&1&0&0\\0&1&0&1&0\\0&2&0&-2&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0\\0&0&1&0&0\\0&1&0&1&0\\0&0&0&-4&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0\\0&1&0&1&0\\0&0&1&0&0\\0&0&0&-4&0\end{bmatrix}$

จะได้ว่า $c_1=c_2=c_3=c_4=0$ เป็น linearly independent

ทำให้ได้ว่า $S_2 = S_2'$

- พิจารณา $S_3' = \left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}\right\}$

จะได้ว่า $c_1\begin{bmatrix}1\\0\\-1\\2\end{bmatrix} + c_2\begin{bmatrix}0\\0\\1\\2\end{bmatrix} + c_3\begin{bmatrix}0\\1\\0\\0\end{bmatrix} + c_4\begin{bmatrix}1\\0\\0\\0\end{bmatrix} + c_5\begin{bmatrix}0\\0\\1\\0\end{bmatrix} = 0$

$\hookrightarrow \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\-1&1&0&0&1&0\\2&2&0&0&0&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&1&1&0\\0&2&0&-2&0&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&1&1&0\\0&0&0&-4&-2&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&1&1&0\\0&0&0&1&\frac{1}{2}&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&0&-\frac{1}{2}&0\\0&0&1&0&0&0\\0&1&0&0&\frac{1}{2}&0\\0&0&0&1&\frac{1}{2}&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&0&-\frac{1}{2}&0\\0&1&0&0&\frac{1}{2}&0\\0&0&1&0&0&0\\0&0&0&1&\frac{1}{2}&0\end{bmatrix}$

จะได้เป็น $\begin{aligned}c_1 - \tfrac{1}{2}c_5 &= 0\\ c_2 + \tfrac{1}{2}c_5 &= 0\\ c_3 &= 0\\ c_4 + \tfrac{1}{2}c_5 &= 0\end{aligned} \Rightarrow \begin{aligned}c_1 &= \tfrac{1}{2}c_5\\ c_2 &= -\tfrac{1}{2}c_5\\ c_4 &= -\tfrac{1}{2}c_5\end{aligned}$ โดยที่ c_5 เป็น free variable ; $c_5 \in \mathbb{R}$

จะได้ $c = c_5\begin{bmatrix}\frac{1}{2}\\-\frac{1}{2}\\0\\-\frac{1}{2}\\1\end{bmatrix} + c_5\begin{bmatrix}0\\0\\1\\0\\0\end{bmatrix} \rightarrow$ ไม่เป็น linearly independent

ทำให้ได้ว่า $S_3 = S_2$

- พิจารณา $S_4' = \left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right\}$

จะได้ว่า $c_1\begin{bmatrix}1\\0\\-1\\2\end{bmatrix} + c_2\begin{bmatrix}0\\0\\1\\2\end{bmatrix} + c_3\begin{bmatrix}0\\1\\0\\0\end{bmatrix} + c_4\begin{bmatrix}1\\0\\0\\0\end{bmatrix} + c_5\begin{bmatrix}0\\0\\0\\1\end{bmatrix} = 0$

$\hookrightarrow \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\-1&1&0&0&0&0\\2&2&0&0&1&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&1&0&0\\0&2&0&-2&1&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&1&0&0\\0&0&0&-4&1&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&1&0&0\\0&0&1&0&0&0\\0&1&0&1&0&0\\0&0&0&1&-\frac{1}{4}&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&0&\frac{1}{4}&0\\0&0&1&0&0&0\\0&1&0&0&\frac{1}{4}&0\\0&0&0&1&-\frac{1}{4}&0\end{bmatrix} \sim \begin{bmatrix}1&0&0&0&\frac{1}{4}&0\\0&1&0&0&\frac{1}{4}&0\\0&0&1&0&0&0\\0&0&0&1&-\frac{1}{4}&0\end{bmatrix}$

จะได้เป็น $\begin{aligned}c_1 + \tfrac{1}{4}c_5 &= 0\\ c_2 + \tfrac{1}{4}c_5 &= 0\\ c_3 &= 0\\ c_4 - \tfrac{1}{4}c_5 &= 0\end{aligned} \Rightarrow \begin{aligned}c_1 &= -\tfrac{1}{4}c_5\\ c_2 &= -\tfrac{1}{4}c_5\\ c_4 &= \tfrac{1}{4}c_5\end{aligned}$ โดยที่ c_5 เป็น free variable ; $c_5 \in \mathbb{R}$

จะได้ $c = c_5\begin{bmatrix}-\frac{1}{4}\\-\frac{1}{4}\\0\\\frac{1}{4}\\1\end{bmatrix} + c_5\begin{bmatrix}0\\0\\1\\0\\0\end{bmatrix} \rightarrow$ ไม่เป็น linearly independent

ทำให้ได้ว่า $S_4 = S_2$

$\therefore$ Basis ของ $\mathbb{R}^4$ คือ $\left\{\begin{bmatrix}1\\0\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\0\\0\end{bmatrix}\right\}$ // Ans.

9. Find the coordinate vector for $\begin{bmatrix}1\\2\\3\end{bmatrix}$ relative to the ordered basis

$$\left\{\begin{bmatrix}1\\1\\3\end{bmatrix},\begin{bmatrix}1\\2\\3\end{bmatrix},\begin{bmatrix}1\\0\\1\end{bmatrix}\right\}.$$

Sol: จากโจทย์ ให้ $v=\begin{bmatrix}1\\2\\3\end{bmatrix}$ และ $\beta=\left\{\begin{bmatrix}1\\1\\3\end{bmatrix},\begin{bmatrix}1\\2\\3\end{bmatrix},\begin{bmatrix}1\\0\\1\end{bmatrix}\right\}$

ต้องการหา $[v]_\beta$

ให้ $v=x_1\begin{bmatrix}1\\1\\3\end{bmatrix}+x_2\begin{bmatrix}1\\2\\3\end{bmatrix}+x_3\begin{bmatrix}1\\0\\1\end{bmatrix}$ จะได้ว่า $\begin{aligned}x_1+x_2+x_3&=1\\x_1+2x_2&=2\\3x_1+3x_2+x_3&=3\end{aligned}$

ได้เป็น $\left[\begin{array}{ccc|c}1&1&1&1\\1&2&0&2\\3&3&1&3\end{array}\right]\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&-1&1\\0&0&-2&0\end{array}\right]\begin{matrix}\sim R_2-R_1\\\sim R_3-3R_1\end{matrix}\sim\left[\begin{array}{ccc|c}1&0&2&0\\0&1&-1&1\\0&0&1&0\end{array}\right]\begin{matrix}\sim R_1-R_2\\\sim -\frac{1}{2}R_3\end{matrix}\sim\left[\begin{array}{ccc|c}1&0&0&0\\0&1&0&1\\0&0&1&0\end{array}\right]\begin{matrix}\sim R_1-2R_3\\\sim R_2+R_3\end{matrix}$

ทำให้ได้ว่า $x_1=0,\ x_2=1,\ x_3=0$

นั่นคือ $[v]_\beta=\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$

$\therefore\ [v]_\beta=\begin{bmatrix}0\\1\\0\end{bmatrix}$ # Ans.

10. Compute the rank and nullity of

$$A=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}.$$

Sol: จากโจทย์ จะได้ว่า $\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix}\sim\begin{bmatrix}1&2&3\\0&-3&-6\\0&-6&-12\end{bmatrix}\begin{matrix}\sim R_2-4R_1\\\sim R_3-7R_1\end{matrix}$

$\sim\begin{bmatrix}1&2&3\\0&-3&-6\\0&0&0\end{bmatrix}\sim R_3-2R_2$

$\sim\begin{bmatrix}1&2&3\\0&1&2\\0&0&0\end{bmatrix}\sim -\frac{1}{3}R_2$

ทำให้ได้ว่า $\text{rank}(A)=2$

และจาก $\text{rank}(A)+\text{nullity}(A)=n(\text{Col})$

ทำให้ได้ว่า $\begin{aligned}\text{nullity}(A)&=n(\text{Col})-\text{rank}(A)\\&=3-2\\&=1\end{aligned}$

$\therefore\ \text{rank}(A)=2$ และ $\text{nullity}(A)=1$ # Ans.